

The empirical recurrence for OEIS A250531, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A250531 records “Empirical recurrence of order 65” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 121$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 67$. Its last coefficient is $c_{65} = -144 \neq 0$, so no further tail satisfies anything shorter; any order-65 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A250531 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with nondecreasing $\max(x(i,j), x(i,j-1))$ in the i direction and nondecreasing $\min(x(i,j), x(i-1,j))$ in the j direction.”. It has offset 1 and begins

3958, 44017, 393910, 3091263, 21783369, 142934266, 885728348, 5269466799, ...

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:30:37, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 121$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 121$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 65: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 65.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 242$ exact terms from $n = 2$ on, it returns order 65 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	30	c_{18}	−119428179490	c_{35}	−29425980781649	c_{52}	−16242733639
c_2	−351	c_{19}	242279835009	c_{36}	18918486258718	c_{53}	−521984200
c_3	1631	c_{20}	179218378873	c_{37}	16178028531659	c_{54}	2637928684
c_4	3375	c_{21}	−1017780457031	c_{38}	−20488946212718	c_{55}	−394887363
c_5	−74696	c_{22}	439682830927	c_{39}	−3425954556145	c_{56}	−253215647
c_6	278460	c_{23}	2470244047723	c_{40}	14075016338443	c_{57}	82433372
c_7	427194	c_{24}	−3249041241432	c_{41}	−2804555179800	c_{58}	10402223
c_8	−6404307	c_{25}	−3167297213085	c_{42}	−6489063025637	c_{59}	−7903238
c_9	13673814	c_{26}	9338992676745	c_{43}	3347462362275	c_{60}	420475
c_{10}	42946891	c_{27}	−471299933438	c_{44}	1859641219444	c_{61}	366205
c_{11}	−254331829	c_{28}	−16350911144539	c_{45}	−1859886181584	c_{62}	−63663
c_{12}	156815154	c_{29}	11129167395472	c_{46}	−177819536582	c_{63}	−4432
c_{13}	1796307546	c_{30}	17793905238513	c_{47}	660269498052	c_{64}	1992
c_{14}	−4426855990	c_{31}	−25323126648076	c_{48}	−108162284260	c_{65}	−144
c_{15}	−4283558726	c_{32}	−8783482912472	c_{49}	−153779409594		
c_{16}	32205772219	c_{33}	33628187020283	c_{50}	60777536718		
c_{17}	−23300648563	c_{34}	−6912991332166	c_{51}	20891948368		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 67$, and it is the recurrence OEIS A250531 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{65} - c_1 x^{65-1} - \dots - c_{65}$. Its constant term is $-c_{65} = 144$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 65 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 65, so $\deg q = 65$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 65, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 65, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -144 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-65 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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